

BIODISC User Manual

Biology Discovery and Intelligence System

Version: 4.7

Publication Date: April 22, 2026

System Type: AGI-Inspired Autonomous Scientific Discovery Framework

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1. Introduction

BIODISC (Biology Discovery and Intelligence System) is an advanced artificial intelligence framework designed specifically for autonomous biological research and scientific discovery. Unlike traditional knowledge retrieval systems, BIODISC actively generates novel hypotheses, designs experiments, and prioritizes research directions.

1.1 What Makes BIODISC Different

Traditional Systems	BIODISC
Retrieves existing knowledge	Creates new knowledge
Answers factual questions	Generates testable hypotheses
Static information	Dynamic causal modeling
Single-scale analysis	Multi-scale integration
Passive response	Active discovery

1.2 Who Should Use BIODISC

- **Biological Researchers:** Explore novel hypotheses and research directions
- **Graduate Students:** Learn complex biological systems through guided discovery
- **Pharmaceutical Scientists:** Identify drug targets and mechanisms
- **Systems Biologists:** Integrate knowledge across scales
- **Clinical Researchers:** Connect molecular mechanisms to disease phenotypes
- **Bioinformatics Analysts:** Interpret high-throughput data in biological context

2. System Overview

2.1 System Architecture

BIODISC consists of three main architectural layers:

1. Bio-Cognitive Discovery Layer (BCDL) The transformative layer that enables knowledge creation through:

- Abductive hypothesis generation
- Temporal causal discovery
- Multi-scale reasoning
- Experimental design
- Research prioritization

2. Domain Knowledge Layer Ten specialized biology domains providing deep expertise:

- Molecular Biology
- Biochemistry
- Genetics
- Cell Biology
- Biophysics
- Bioinformatics
- Computational Biology
- Genomics
- Proteomics
- Systems Biology

3. Core Infrastructure Foundational capabilities including:

- Physics Engine (for biophysical modeling)
- Memory Systems (MORK Ontology, Context Graph, Episodic Memory)
- Causal Discovery Engines
- Meta-Cognitive Capabilities

2.2 System Characteristics

Characteristic	Description
Knowledge Creation	Generates novel scientific hypotheses
Causal Understanding	Learns causal structure from temporal data
Scale Integration	Connects molecular to organism-level phenomena
Experimental Design	Creates protocols optimized for information gain
Research Prioritization	Calculates expected value of research directions
Capability Estimate	70-75% AGI-level functionality

3. Core Capabilities

3.1 Knowledge Retrieval

BIODISC provides comprehensive access to biological knowledge across all domains:

Example Query:

"What is the central dogma of molecular biology?"

BIODISC provides a detailed explanation covering DNA transcription, RNA translation, and protein synthesis, with connections to related concepts in genetics and biochemistry.

3.2 Cross-Domain Synthesis

BIODISC integrates knowledge across multiple biology domains:

Example Query:

"How do genetic mutations affect protein structure and cellular function?"

BIODISC synthesizes information from genetics (DNA mutations), molecular biology (protein structure), and cell biology (cellular function) to provide a comprehensive answer.

3.3 Causal Reasoning

BIODISC can reason about causal relationships in biological systems:

Example Query:

"What causes protein misfolding in neurodegenerative diseases?"

BIODISC identifies causal factors, mechanisms, and downstream consequences.

4. The Bio-Cognitive Discovery Layer

The Bio-Cognitive Discovery Layer (BCDL) is BIODISC's transformative capability that enables it to transition from knowledge retrieval to knowledge creation.

4.1 Module 1: Abductive Theory Former

Generates novel scientific hypotheses through abductive reasoning (inference to best explanation).

Capabilities:

- Generate novel hypotheses about biological phenomena
- Identify gaps in current knowledge
- Synthesize coherent theories from disparate findings
- Propose mechanistic explanations

Example Query:

"Generate hypotheses about why certain protein aggregates are toxic in Alzheimer's disease while others are not."

BIODISC generates multiple testable hypotheses, each with:

- Clear statement of the hypothesis
- Supporting evidence and reasoning
- Novelty assessment
- Suggested experimental validation

4.2 Module 2: Temporal Causal Discovery Engine

Learns causal structure from time-series biological data and models dynamic biological processes.

Capabilities:

- Learn causal graphs from temporal data (gene expression, live imaging, etc.)
- Model biological processes as dynamic systems
- Predict future system states
- Identify optimal intervention points

Example Query:

"Analyze this time-series gene expression data to identify which genes regulate each other during cellular stress response."

BIODISC applies Granger causality and other methods to identify regulatory relationships and predict network dynamics.

4.3 Module 3: Multi-Scale Reasoner

Integrates knowledge across biological scales from molecular to organismal levels.

Capabilities:

- Map entities across scales (molecules → cells → tissues → organs → organisms)
- Discover emergent properties that arise at specific scales
- Model how perturbations propagate across scales
- Identify scale-invariant mechanisms

Example Query:

"How do mitochondrial DNA mutations at the molecular scale affect tissue-level metabolism and organismal health?"

BIODISC traces the causal chain across scales, identifying emergent properties and potential intervention points.

4.4 Module 5: Discovery Value Calculator

Prioritizes research directions using information theory and calculates expected value of discoveries.

Capabilities:

- Prioritize research hypotheses by expected information gain
- Calculate scientific impact predictions
- Optimize resource allocation across projects
- Identify high-value research opportunities

Example Query:

"Given my lab's expertise in proteomics and a budget of \$150,000, which research direction on protein aggregation would yield the highest scientific impact?"

BIODISC provides ranked research priorities with expected value, risk assessment, and resource allocation recommendations.

5. Using Discovery Mode

Discovery mode is BIODISC's most powerful capability, enabling autonomous scientific discovery through the integrated use of all five BCDL modules.

5.1 When to Use Discovery Mode

Use discovery mode when you want to:

- Explore novel research directions
- Generate hypotheses for unexplained phenomena
- Design experiments to test ideas
- Prioritize research investments
- Understand multi-scale biological systems
- Learn causal structure from your data

5.2 Discovery Mode Workflow

When you ask a discovery-mode question, BIODISC:

1. **Analyzes your query** to understand the biological phenomenon 2. **Generates hypotheses** using abductive reasoning 3. **Identifies knowledge gaps** in current understanding 4. **Searches for causal mechanisms** across scales 5. **Designs experiments** to test the best hypotheses 6. **Prioritizes research** based on expected value 7. **Synthesizes findings** into a coherent theory

5.3 Natural Language Query Examples

Hypothesis Generation:

"Generate hypotheses about the molecular mechanisms underlying the relationship between gut microbiome composition and brain function."

Mechanism Exploration:

"What are the possible mechanisms by which autophagy regulates protein aggregate clearance in neurodegenerative diseases?"

Experimental Design:

"Design experiments to test whether protein X aggregation is a cause or consequence of cellular toxicity in Huntington's disease models."

Multi-Scale Analysis:

"How do epigenetic modifications at the DNA level translate into changes in cellular behavior and tissue pathology in cancer development?"

Causal Discovery:

"Analyze how signaling pathways interact during cellular differentiation and identify key regulatory nodes."

Research Prioritization:

"What are the most promising research directions for understanding drug resistance in cancer therapy, given limited resources?"

6. Biology Domains

BIODISC includes ten specialized biology domains that work together to provide comprehensive coverage of biological knowledge.

6.1 Molecular Biology

Focus: DNA replication, transcription, translation, gene expression regulation

Example Query:

"Explain the molecular mechanisms of transcriptional regulation in eukaryotic cells."

6.2 Biochemistry

Focus: Chemical processes within living organisms, enzyme kinetics, metabolic pathways

Example Query:

"Describe the biochemical pathways involved in ATP synthesis during cellular respiration."

6.3 Genetics

Focus: Heredity, genetic variation, gene function, inheritance patterns

Example Query:

"How do single nucleotide polymorphisms contribute to genetic disease susceptibility?"

6.4 Cell Biology

Focus: Cell structure, function, organelles, cell cycle, cell signaling

Example Query:

"What are the molecular mechanisms that control the cell cycle checkpoints?"

6.5 Biophysics

Focus: Physical principles of biological systems, protein folding, membrane dynamics

Example Query:

"What physical forces drive protein folding and how do mutations affect stability?"

6.6 Bioinformatics

Focus: Analysis of biological data, algorithms, statistical methods

Example Query:

"How can we identify regulatory motifs in promoter sequences using bioinformatics approaches?"

6.7 Computational Biology

Focus: Mathematical and computational modeling of biological systems

Example Query:

"How do computational models help us understand biological network dynamics?"

6.8 Genomics

Focus: Genome structure, function, evolution, comparative genomics

Example Query:

"What have genome-wide association studies revealed about complex disease architecture?"

6.9 Proteomics

Focus: Protein structure, function, interactions, modifications

Example Query:

"How does post-translational modification regulate protein function in signaling networks?"

6.10 Systems Biology

Focus: Integration of biological components, networks, emergent properties

Example Query:

"How do systems biology approaches reveal emergent properties in metabolic networks?"

7. Advanced Features

7.1 Meta-Context Engine

Provides layered context understanding for complex queries, considering:

- Temporal context (how understanding changes over time)
- Perceptual context (different perspectives on the same phenomenon)
- Theoretical context (competing scientific frameworks)
- Experimental context (how data shapes understanding)

7.2 Memory Systems

MORK Ontology: Hierarchical knowledge organization

Context Graph: Relationships between concepts

Working Memory: Active reasoning (7+/-2 items)

Episodic Memory: Learning from previous queries

7.3 Physics Engine

Models biophysical processes including:

- Molecular dynamics

- Thermodynamics
- Reaction kinetics
- Diffusion processes

8. Example Discovery Sessions

Example Session 1: Neurodegenerative Disease Research

Researcher Query:

"I'm studying protein misfolding in Alzheimer's disease. What are the key mechanisms, and what experiments would help distinguish between competing theories?"

BIODISC Discovery Process:

Step 1 - Hypothesis Generation: BIODISC generates five novel hypotheses: 1. Amyloid-beta oligomers disrupt synaptic function through membrane receptor interactions 2. Tau protein propagation follows a prion-like spreading mechanism 3. Impaired proteostasis network is the primary driver of aggregation 4. Neuroinflammation creates a permissive environment for aggregate formation 5. Metal ion dyshomeostasis triggers protein conformational changes

Step 2 - Knowledge Gap Analysis: BIODISC identifies gaps:

- Relative contribution of each mechanism in different disease stages
- Cell-type specific vulnerability factors
- Temporal ordering of pathological events

Step 3 - Multi-Scale Analysis: BIODISC maps mechanisms across scales:

- Molecular: Protein conformational changes
- Cellular: Organelle dysfunction and stress responses
- Tissue: Synaptic loss and network disruption
- Organism: Cognitive decline patterns

Step 4 - Experimental Design: BIODISC designs experiments:

- Primary experiment: Use induced neurons with isogenic controls to test oligomer toxicity
- Controls: Non-toxic oligomers, scrambled sequences, vehicle controls
- Predicted outcomes: Dose-response relationship with specific threshold effects
- Information gain: High - distinguishes between receptor-mediated vs. direct toxicity mechanisms

Step 5 - Research Prioritization: BIODISC prioritizes research directions based on:

- Information gain (understanding mechanism)
- Impact (potential therapeutic targets)
- Feasibility (experimental accessibility)
- Resource efficiency (cost vs. knowledge gained)

Example Session 2: Cancer Therapy Resistance

Researcher Query:

"My lab studies targeted therapy resistance in cancer. Help me understand why patients develop resistance and suggest promising research directions."

BIODISC Discovery Process:

Step 1 - Hypothesis Generation: BIODISC generates hypotheses about resistance mechanisms: 1. Pre-existing resistant subclones expand under selective pressure 2. Cancer cells activate alternative signaling pathways 3. Phenotypic plasticity enables lineage switching 4. Tumor microenvironment provides protective signals 5. Epigenetic reprogramming rewires cellular identity

Step 2 - Temporal Causal Analysis: For time-series gene expression data during treatment, BIODISC identifies:

- Early response genes (within hours)
- Adaptive changes (days to weeks)
- Resistance-associated pathways (long-term)

Step 3 - Multi-Scale Integration: BIODISC connects:

- Molecular: Mutation and expression changes
- Cellular: Survival pathway activation
- Tissue: Microenvironment interactions
- Organism: Clinical progression patterns

Step 4 - Experimental Design: BIODISC designs a longitudinal study:

- Collect samples at multiple time points during treatment
- Use single-cell sequencing to capture heterogeneity
- Apply spatial transcriptomics to map microenvironment
- Predict optimal sampling intervals for maximal information gain

Step 5 - Research Prioritization: BIODISC recommends focusing on early adaptive changes, as interventions at this stage may prevent resistance development.

Example Session 3: Synthetic Biology Design

Researcher Query:

"I want to design a genetic circuit for biosensing environmental toxins. What are the key considerations, and how can I optimize the design?"

BIODISC Discovery Process:

Step 1 - Knowledge Synthesis: BIODISC integrates knowledge from:

- Molecular biology (promoters, ribosome binding sites)
- Systems biology (circuit dynamics, feedback)
- Biophysics (diffusion, binding kinetics)
- Computational biology (modeling and simulation)

Step 2 - Hypothesis Generation: BIODISC generates design hypotheses:

- Use positive feedback for signal amplification
- Implement threshold detection via ultrasensitive circuits
- Add orthogonal regulation for tunability
- Include noise suppression mechanisms

Step 3 - Multi-Scale Analysis: BIODISC considers:

- Molecular: DNA sequence optimization
- Cellular: Metabolic burden, resource allocation
- Population: Cell-cell communication
- Environmental: Toxin diffusion and stability

Step 4 - Experimental Design: BIODISC proposes:

- Characterize individual components first
- Test circuit in controlled conditions
- Vary environmental parameters systematically
- Use flow cytometry for single-cell resolution

Step 5 - Value Calculation: BIODISC prioritizes experiments that maximize information gain about circuit behavior while minimizing resource use.

9. Best Practices

9.1 Formulating Effective Queries

Be Specific About Your Question:

- Good: "What molecular mechanisms regulate protein folding in the endoplasmic reticulum?"
- Less effective: "Tell me about protein folding"

Provide Context When Available:

- "Given that mutations in protein X cause misfolding, what chaperone systems might be involved?"

Ask About Mechanisms:

- "How does..." questions often yield deeper insights than "What..." questions

Invite Hypothesis Generation:

- "Generate hypotheses about..." explicitly engages BIODISC's discovery capabilities

9.2 Interpreting Results

Hypothesis Novelty: BIODISC provides novelty scores. Higher scores indicate more original hypotheses.

Confidence Levels: BIODISC indicates confidence in each answer. Lower confidence may indicate areas needing more research.

Multi-Domain Connections: Pay attention to cross-domain insights, as these often reveal novel connections.

9.3 Iterative Discovery

Refine Hypotheses: Use BIODISC to refine initial hypotheses based on new information.

Design Follow-up Experiments: Ask BIODISC to design experiments based on previous results.

Update Knowledge: BIODISC learns from the research context you provide.

10. Technical Reference

10.1 System Requirements

Minimum:

- Python 3.8+
- 8GB RAM
- 1GB free disk space

Recommended:

- Python 3.10+
- 16GB RAM
- 4GB free disk space
- NumPy, SciPy for numerical computations

10.2 Accessing BIODISC

Python API: `python from biodisc_core import create_biodisc_system`

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```
system = create_biodisc_system()
```

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```
answer = system.answer("Your question here")
```

Discovery Mode: `python from biodisc_core.bio_cognitive import BioCognitiveDiscoveryLayer`

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```
biodisc = BioCognitiveDiscoveryLayer({})
```

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```
results = biodisc.discover("Your discovery question") ``
```

10.3 Domain-Specific Access

Access specific biology domains:

```
``python from biodisc_core.domains import DomainRegistry
registry = DomainRegistry() registry.auto_load_domains()
```

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```
result = registry.process_query("Your question") ``
```

10.4 Capability Detection

Check available capabilities:

```
``python from biodisc_core.bio_cognitive import ( ABDUCTIVE_THEORY_FORMER_AVAILABLE,
TEMPORAL_CAUSAL_DISCOVERY_AVAILABLE,          MULTI_SCALE_REASONER_AVAILABLE,
EXPERIMENTAL_DESIGNER_AVAILABLE, DISCOVERY_VALUE_CALCULATOR_AVAILABLE )

print(f"Abductive Theory Former: {ABDUCTIVE_THEORY_FORMER_AVAILABLE}") print(f"Temporal Causal
Discovery:      {TEMPORAL_CAUSAL_DISCOVERY_AVAILABLE}")      print(f"Multi-Scale      Reasoner:
{MULTI_SCALE_REASONER_AVAILABLE}")                          print(f"Experimental      Designer:
{EXPERIMENTAL_DESIGNER_AVAILABLE}")                          print(f"Discovery      Value      Calculator:
{DISCOVERY_VALUE_CALCULATOR_AVAILABLE}") ``
---
```

Appendix A: Example Query Catalog

Molecular Biology

- "How do epigenetic modifications affect gene expression during cellular differentiation?"
- "What are the molecular mechanisms of DNA damage repair?"
- "How does alternative splicing generate protein diversity?"

Cell Biology

- "What controls the transition from cell growth to cell division?"
- "How do cells maintain organelle identity during membrane trafficking?"
- "What mechanisms regulate programmed cell death?"

Genetics

- "How do genetic variants contribute to complex trait variation?"
- "What are the patterns of inheritance for mitochondrial disorders?"
- "How do gene-gene interactions (epistasis) affect phenotypes?"

Biochemistry

- "How do allosteric regulators control enzyme activity?"
- "What are the thermodynamic constraints on metabolic pathway efficiency?"
- "How do cells maintain redox homeostasis?"

Systems Biology

- "What emergent properties arise in protein interaction networks?"
- "How do feedback loops control biological oscillators?"
- "What principles govern biological network robustness?"

Neuroscience

- "How do synaptic changes underlie learning and memory?"
- "What neural circuits regulate sleep-wake cycles?"
- "How does the brain integrate multisensory information?"

Cancer Biology

- "What are the hallmarks of cancer and how do they interact?"
- "How does the tumor microenvironment influence metastasis?"
- "What mechanisms underlie therapy resistance?"

Immunology

- "How does the immune system distinguish self from non-self?"
- "What are the mechanisms of immunological memory?"
- "How does chronic inflammation contribute to disease?"

Developmental Biology

- "How do morphogen gradients pattern embryonic tissues?"
- "What mechanisms control stem cell differentiation?"
- "How do organisms scale organ size with body size?"

Appendix B: Discovery Mode Templates

Template 1: Mechanism Discovery

"What are the possible mechanisms by which [X] affects [Y] in [biological system]?"

Example: "What are the possible mechanisms by which gut microbiota affects brain function in neurodegenerative disease?"

Template 2: Hypothesis Generation

"Generate hypotheses about [phenomenon] that could explain [observation]."

Example: "Generate hypotheses about protein aggregation that could explain selective neuronal vulnerability in Parkinson's disease."

Template 3: Experimental Design

"Design experiments to distinguish between [hypothesis A] and [hypothesis B] in [system]."

Example: "Design experiments to distinguish whether inflammation causes neurodegeneration or is a secondary response."

Template 4: Multi-Scale Analysis

"How do [molecular-level changes] affect [cellular/tissue/organism-level outcomes]?"

Example: "How do chromatin modifications at specific gene loci affect tissue-level patterns of gene expression during development?"

Template 5: Causal Discovery

"Given [temporal data], what causal relationships can be inferred about [process]?"

Example: "Given time-series gene expression data during drug treatment, what causal relationships can be inferred about resistance mechanisms?"

Template 6: Research Prioritization

"Given [constraints], what are the most promising research directions for [topic]?"

Example: "Given expertise in proteomics and limited funding, what are the most promising research directions for understanding protein aggregation diseases?"

Appendix C: Glossary

Abductive Reasoning: Inference to the best explanation, used for hypothesis generation.

Causal Graph: A graphical representation of causal relationships between variables.

Discovery Mode: BIODISC's mode of operation that generates new knowledge rather than retrieving existing information.

Emergent Property: A property that arises at a specific scale of organization but is not predictable from lower-scale components.

Information Gain: The reduction in uncertainty about a phenomenon resulting from an experiment or observation.

Knowledge Gap: An area where current scientific understanding is incomplete or absent.

Multi-Scale Reasoning: Integration of knowledge across different levels of biological organization (molecular, cellular, tissue, organismal).

Novelty Score: A measure of how original a hypothesis is relative to existing literature.

Research Prioritization: Ranking research directions by expected value, considering scientific impact, feasibility, and resource requirements.

Scale-Invariant Mechanism: A mechanism that operates similarly across different scales of organization.

Temporal Causal Discovery: Learning causal relationships from time-series data.

Document Information

Title: BIODISC User Manual

Version: 4.7

Date: April 22, 2026

Authors: BIODISC Development Team

System: Biology Discovery and Intelligence System

Capability Level: 70-75% AGI-Equivalent

This user manual provides comprehensive guidance for using BIODISC in discovery mode. For technical documentation, API references, and developer guides, please refer to the additional documentation in the biodisc_core directory.